

Examiner  
only

(iii) Use the graph to answer the following questions.

I. Give the year when **equal** numbers of plastic bags and paper bags were used. [1]

Year .....

II. Circle the **simplest** ratio that compares the use of paper bags to plastic bags in 2000. [1]

2 : 8      1 : 4      4 : 1      8 : 2      20 : 80

(iv) Give **one** reason that explains the change seen in plastic bag usage in Wales between 2008 and 2018. [1]

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|    |
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END OF PAPER



### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                   | NEGATIVE IONS |                    |
|---------------|-------------------|---------------|--------------------|
| Name          | Formula           | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$  | bromide       | $\text{Br}^{-}$    |
| ammonium      | $\text{NH}_4^{+}$ | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$  | chloride      | $\text{Cl}^{-}$    |
| calcium       | $\text{Ca}^{2+}$  | fluoride      | $\text{F}^{-}$     |
| copper(II)    | $\text{Cu}^{2+}$  | hydroxide     | $\text{OH}^{-}$    |
| hydrogen      | $\text{H}^{+}$    | iodide        | $\text{I}^{-}$     |
| iron(II)      | $\text{Fe}^{2+}$  | nitrate       | $\text{NO}_3^{-}$  |
| iron(III)     | $\text{Fe}^{3+}$  | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^{+}$   | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$  |               |                    |
| nickel        | $\text{Ni}^{2+}$  |               |                    |
| potassium     | $\text{K}^{+}$    |               |                    |
| silver        | $\text{Ag}^{+}$   |               |                    |
| sodium        | $\text{Na}^{+}$   |               |                    |
| zinc          | $\text{Zn}^{2+}$  |               |                    |



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



28

|          |   |
|----------|---|
| 1        | H |
| Hydrogen | 1 |

|           |    |           |    |
|-----------|----|-----------|----|
| 7         | Li | 9         | Be |
| Lithium   | 3  | Beryllium | 4  |
| 23        | Na | 24        | Mg |
| Sodium    | 11 | Magnesium | 12 |
| 39        | K  | 40        | Ca |
| Potassium | 19 | Calcium   | 20 |
| 86        | Rb | 88        | Sr |
| Rubidium  | 37 | Strontium | 38 |
| 133       | Cs | 137       | Ba |
| Caesium   | 55 | Barium    | 56 |
| 223       | Fr | 226       | Ra |
| Francium  | 87 | Radium    | 88 |
|           |    | 227       | Ac |
|           |    | Actinium  | 89 |

|            |    |           |    |            |    |           |    |          |    |         |    |
|------------|----|-----------|----|------------|----|-----------|----|----------|----|---------|----|
| 11         | B  | 12        | C  | 14         | N  | 16        | O  | 19       | F  | 20      | Ne |
| Boron      | 5  | Carbon    | 6  | Nitrogen   | 7  | Oxygen    | 8  | Fluorine | 9  | Neon    | 10 |
| 27         | Al | 28        | Si | 31         | P  | 32        | S  | 35.5     | Cl | 40      | Ar |
| Aluminium  | 13 | Silicon   | 14 | Phosphorus | 15 | Sulfur    | 16 | Chlorine | 17 | Argon   | 18 |
| 70         | Ga | 73        | Ge | 75         | As | 79        | Se | 80       | Br | 84      | Kr |
| Gallium    | 31 | Germanium | 32 | Arsenic    | 33 | Selenium  | 34 | Bromine  | 35 | Krypton | 36 |
| 115        | In | 119       | Sn | 122        | Sb | 128       | Te | 127      | I  | 131     | Xe |
| Indium     | 49 | Tin       | 50 | Antimony   | 51 | Tellurium | 52 | Iodine   | 53 | Xenon   | 54 |
| 204        | Tl | 207       | Pb | 209        | Bi | 210       | Po | 210      | At | 222     | Rn |
| Thallium   | 81 | Lead      | 82 | Bismuth    | 83 | Polonium  | 84 | Astatine | 85 | Radon   | 86 |
| 59         | Co | 59        | Ni | 63.5       | Cu | 65        | Zn | 65       |    |         |    |
| Cobalt     | 27 | Nickel    | 28 | Copper     | 29 | Zinc      | 30 |          |    |         |    |
| 103        | Rh | 106       | Pd | 108        | Ag | 112       | Cd | 112      |    |         |    |
| Rhodium    | 45 | Palladium | 46 | Silver     | 47 | Cadmium   | 48 |          |    |         |    |
| 192        | Ir | 195       | Pt | 197        | Au | 201       | Hg | 201      |    |         |    |
| Iridium    | 77 | Platinum  | 78 | Gold       | 79 | Mercury   | 80 |          |    |         |    |
| 186        | Re | 188       | Os | 190        |    |           |    |          |    |         |    |
| Rhenium    | 75 |           |    |            |    |           |    |          |    |         |    |
| 184        | W  | 186       | Ta | 188        |    |           |    |          |    |         |    |
| Tungsten   | 74 |           |    |            |    |           |    |          |    |         |    |
| 181        | Hf | 182       | Ti | 183        |    |           |    |          |    |         |    |
| Hafnium    | 72 |           |    |            |    |           |    |          |    |         |    |
| 179        | Y  | 180       | Zr | 181        |    |           |    |          |    |         |    |
| Yttrium    | 39 |           |    |            |    |           |    |          |    |         |    |
| 45         | Sc | 46        | Ti | 47         |    |           |    |          |    |         |    |
| Scandium   | 21 |           |    |            |    |           |    |          |    |         |    |
| 91         | Zr | 92        | Nb | 93         |    |           |    |          |    |         |    |
| Zirconium  | 40 |           |    |            |    |           |    |          |    |         |    |
| 51         | V  | 52        | Cr | 53         |    |           |    |          |    |         |    |
| Vanadium   | 23 | Chromium  | 24 |            |    |           |    |          |    |         |    |
| 55         | Mn | 56        | Fe | 57         |    |           |    |          |    |         |    |
| Manganese  | 25 | Iron      | 26 |            |    |           |    |          |    |         |    |
| 59         | Co | 60        | Ni | 61         |    |           |    |          |    |         |    |
| Cobalt     | 27 | Nickel    | 28 |            |    |           |    |          |    |         |    |
| 101        | Ru | 102       | Rh | 103        |    |           |    |          |    |         |    |
| Ruthenium  | 44 | Rhodium   | 45 |            |    |           |    |          |    |         |    |
| 99         | Tc | 100       | Pd | 101        |    |           |    |          |    |         |    |
| Technetium | 43 | Palladium | 46 |            |    |           |    |          |    |         |    |
| 96         | Mo | 97        | Ag | 98         |    |           |    |          |    |         |    |
| Molybdenum | 42 | Silver    | 47 |            |    |           |    |          |    |         |    |
| 48         | Ti | 49        | V  | 50         |    |           |    |          |    |         |    |
| Titanium   | 22 | Vanadium  | 23 |            |    |           |    |          |    |         |    |
| 45         | Sc | 46        | Cr | 47         |    |           |    |          |    |         |    |
| Scandium   | 21 | Chromium  | 24 |            |    |           |    |          |    |         |    |
| 89         | Y  | 90        | Zr | 91         |    |           |    |          |    |         |    |
| Yttrium    | 39 |           |    |            |    |           |    |          |    |         |    |
| 139        | La | 140       | Ce | 141        |    |           |    |          |    |         |    |
| Lanthanum  | 57 |           |    |            |    |           |    |          |    |         |    |
| 173        | Lu | 174       | Hf | 175        |    |           |    |          |    |         |    |
| Lutetium   | 71 |           |    |            |    |           |    |          |    |         |    |

Key

|        |                      |
|--------|----------------------|
| Ar     | relative atomic mass |
| Symbol |                      |
| Name   |                      |
| Z      | atomic number        |